Guidelines for Human/AI Interaction

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Human/AI Interaction Guidelines

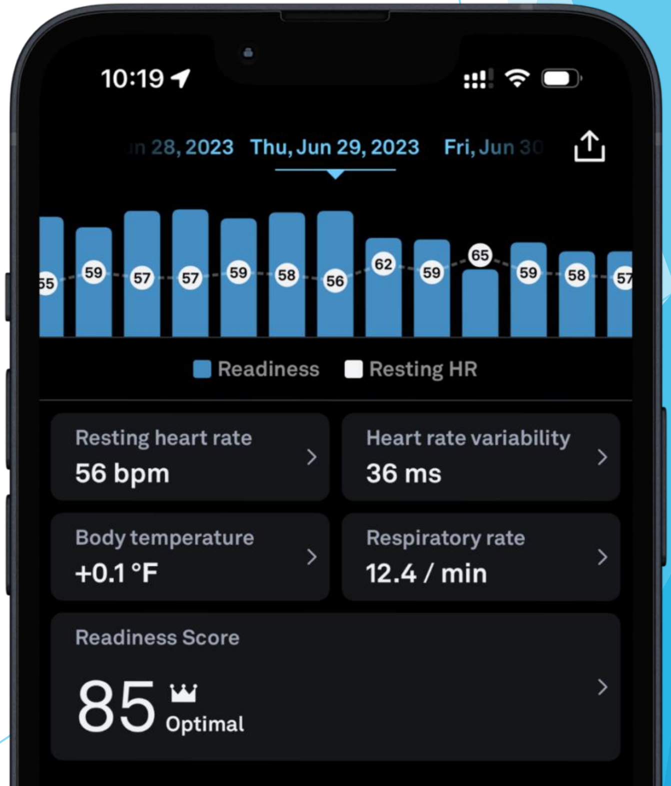
- 18 specific guidelines
  - From 150+ AI design recommendations
  - Validated through three rounds of evaluation
  - Create more human-centric AI-infused systems
- Observable guidelines
  - Not broad guidelines like “build trust” but likely to contribute to building trust
- Ethical concerns
  - Beyond matching of social norms and mitigating social biases
  - Evaluate influences of AI on people and society
  - Can be difficult to evaluate in heuristic evaluation
  - Detection of problems depends on who is performing the evaluation

Interaction with users

- Initially
- During interaction
- When wrong
- Over time

Make clear what the system can do

- Help the user understand what the AI system is capable of doing
- E.g., Oura ring AI assistant will provide wellness tips, answer questions, analyze data
  - Shows all tracked metrics
  - Respiratory rate, heart rate, body temperature, etc.
  - Readiness score for day based on recent activity and sleep

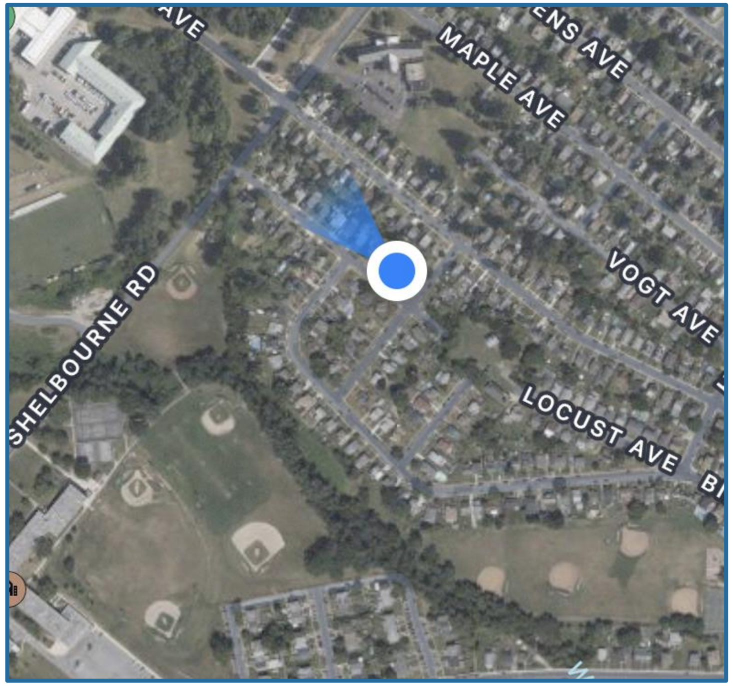


Make clear how well the system can do what it can do

- Help the user understand how often the AI system may make mistakes
- E.g., We *think* you'll love these

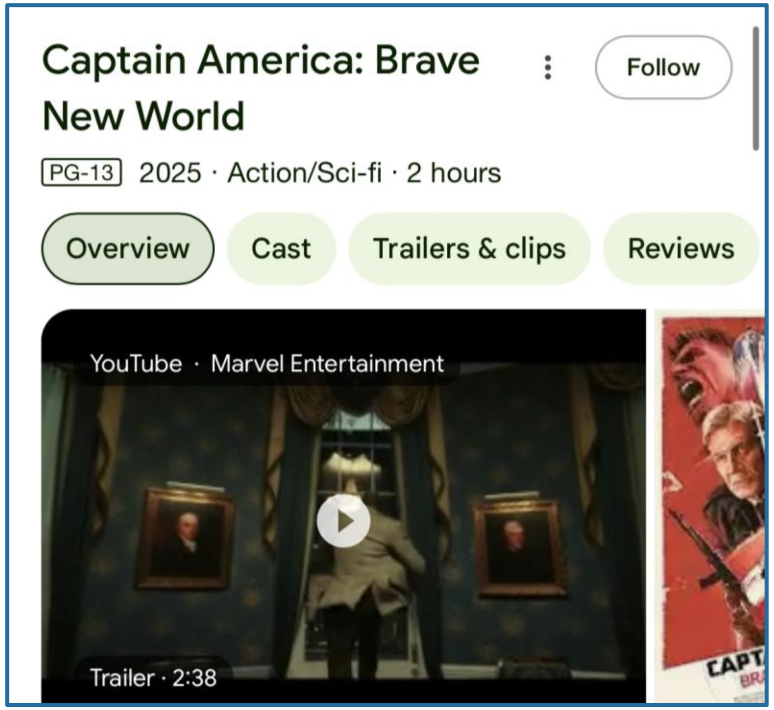
Time services based on context

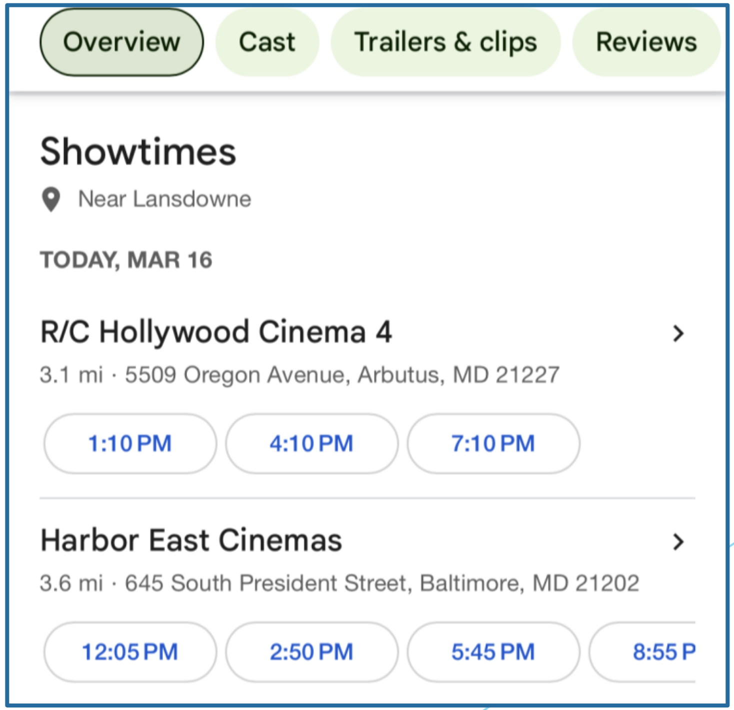
- Time when to act or interrupt based on the user's current task and environment
- E.g., Map updates regularly with location and heading



Show contextually relevant information

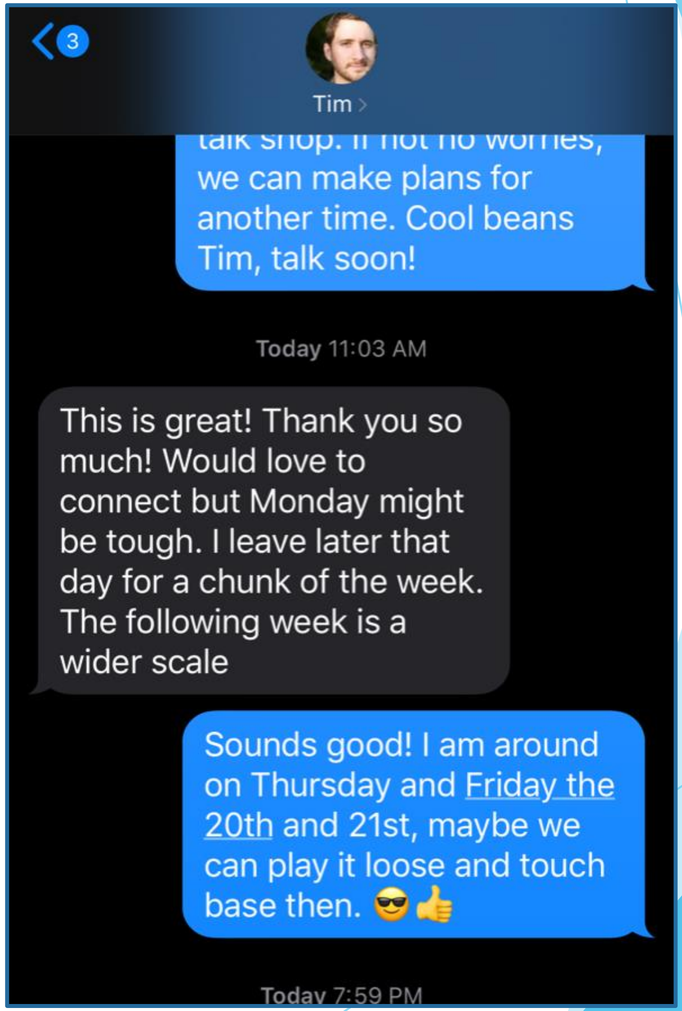
- Display information relevant to the user's current task and environment
- E.g., Google search movie title returns movie theater times nearby





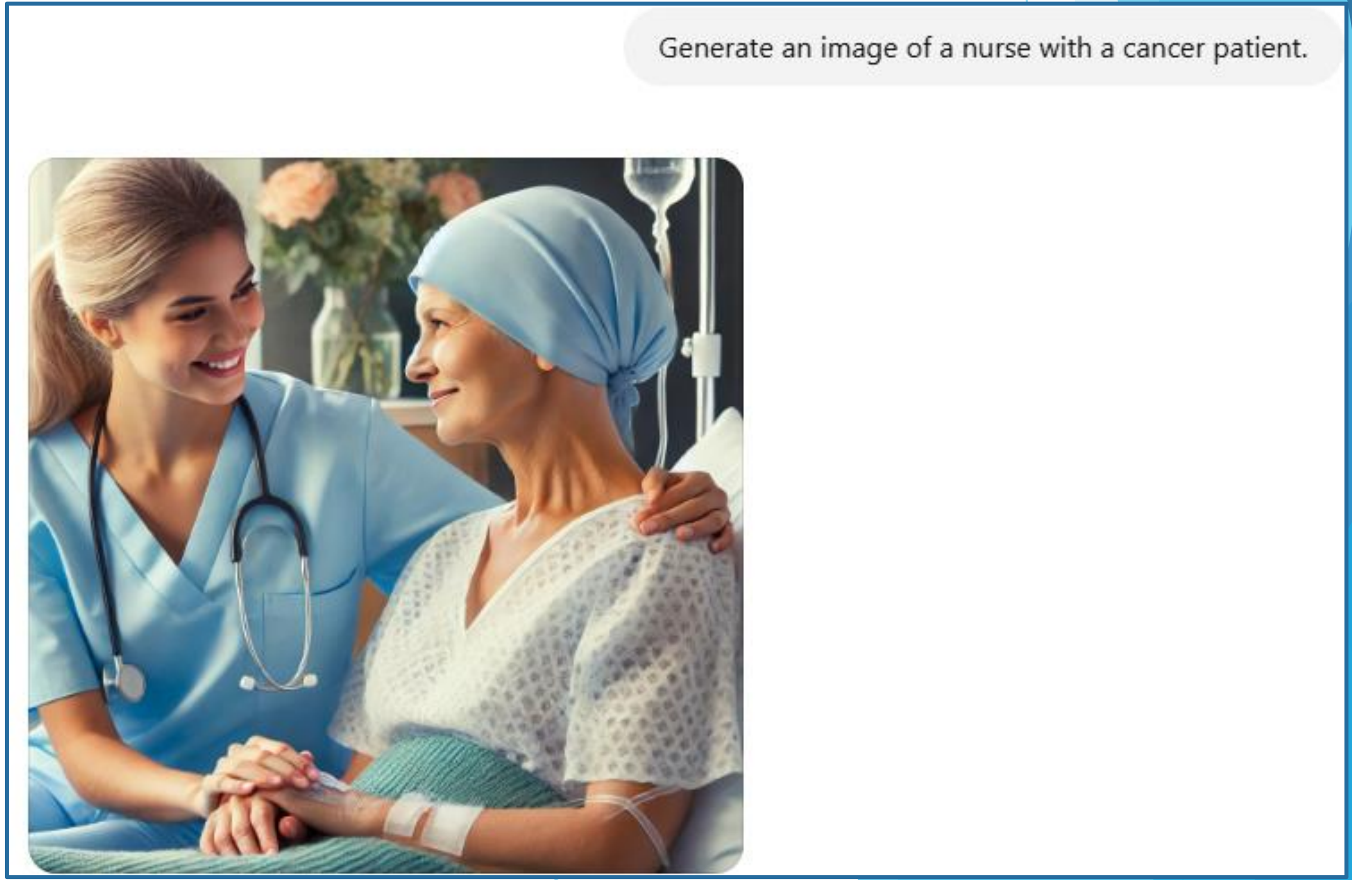
Match relevant social norms

- Ensure the experience is delivered in a way that users would expect, given their social and cultural context
- E.g., Siri reads text abbreviations as plain text



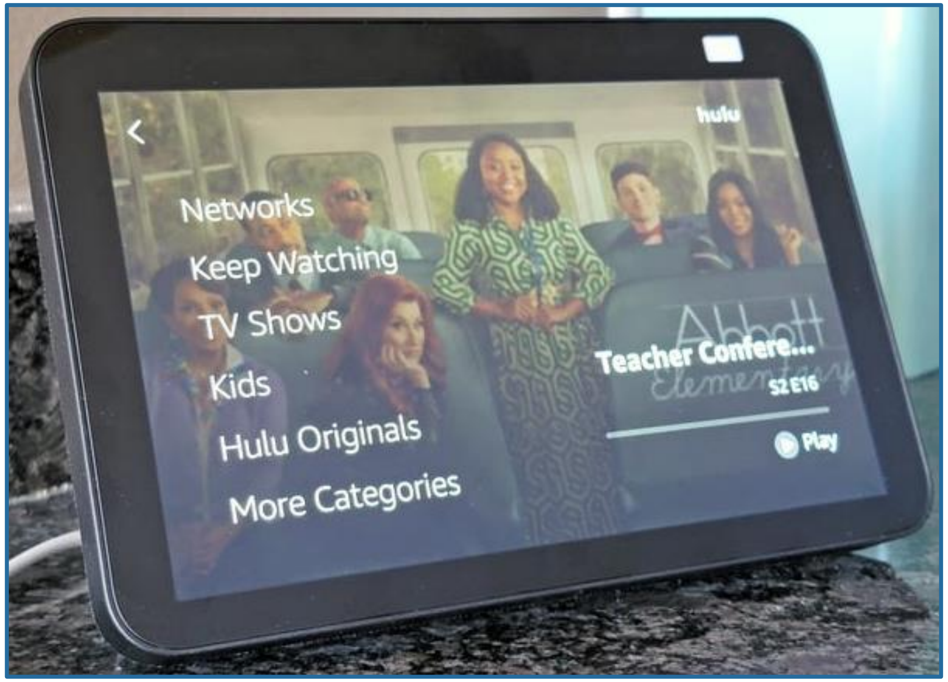
Mitigate social biases

- Ensure the AI system’s language and behaviors do not reinforce undesirable and unfair stereotypes and biases
- E.g., Stereotypes with ChatGPT



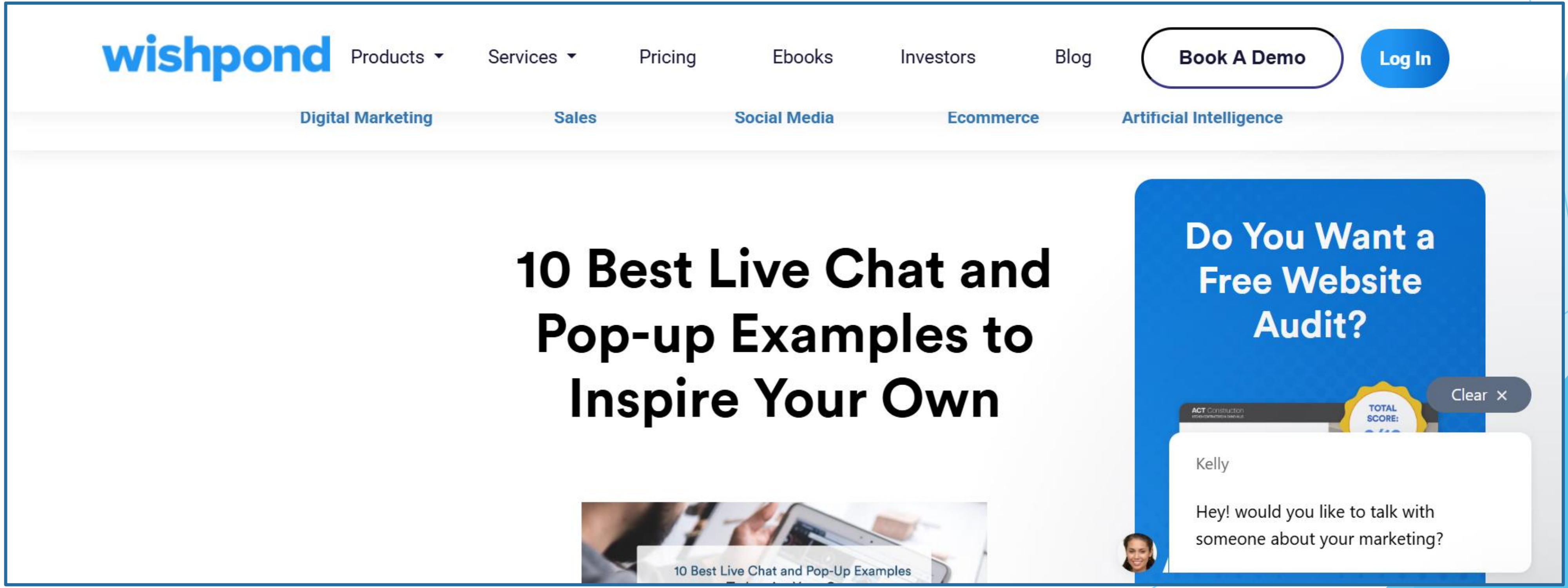
Support efficient invocation

- Make it easy to invoke or request the AI system’s services when needed
- E.g., Can say “Alexa” to wake Amazon devices



Support efficient dismissal

- Make it easy to dismiss or ignore undesired AI system services
- E.g., Chat box can be easily closed



Support efficient correction

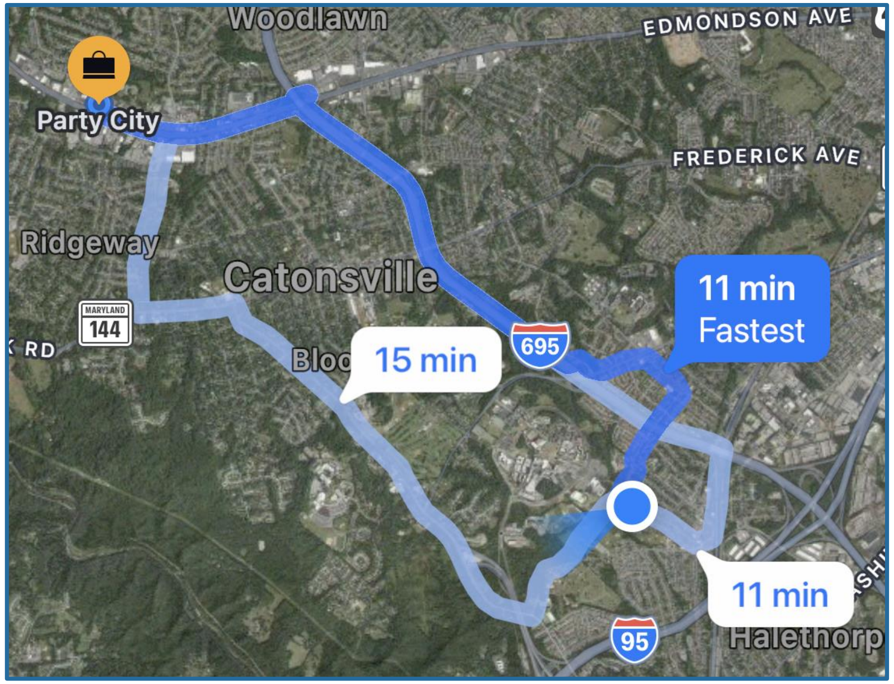
- Make it easy to edit, refine, or recover when the AI system is wrong
- E.g., CarPlay speech to text

Scope services when in doubt

- Engage in disambiguation or gracefully degrade the AI system’s services when uncertain about a user’s goals
- E.g., Multiple suggestions instead of applying directly

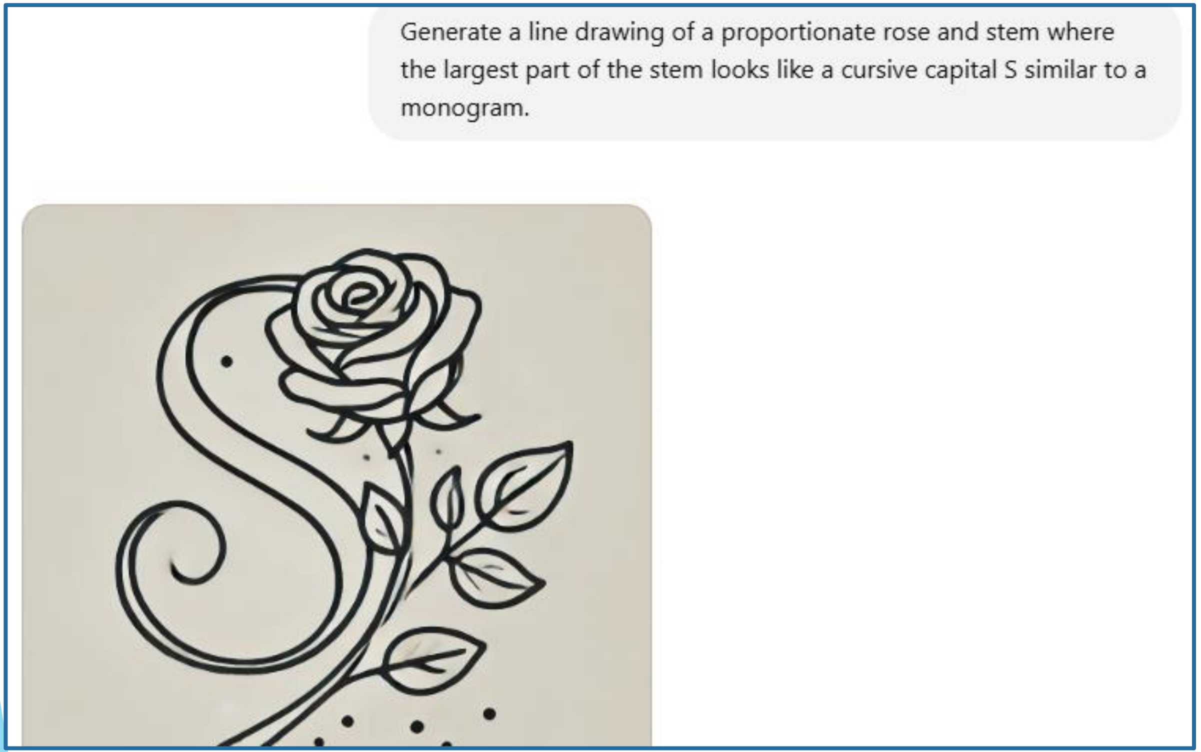
Make clear why the system did what it did

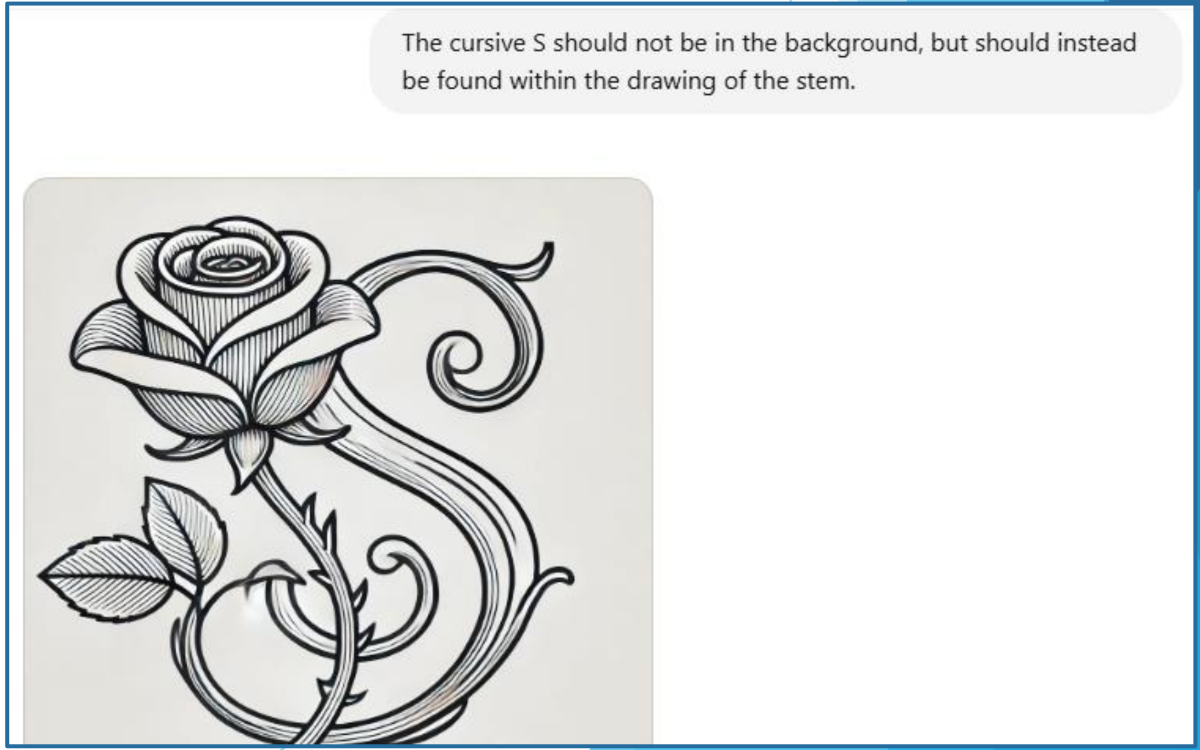
- Enable the user to access an explanation of why the AI system behaved as it did
- E.g., Route selected is labeled as the fastest



Remember recent interactions

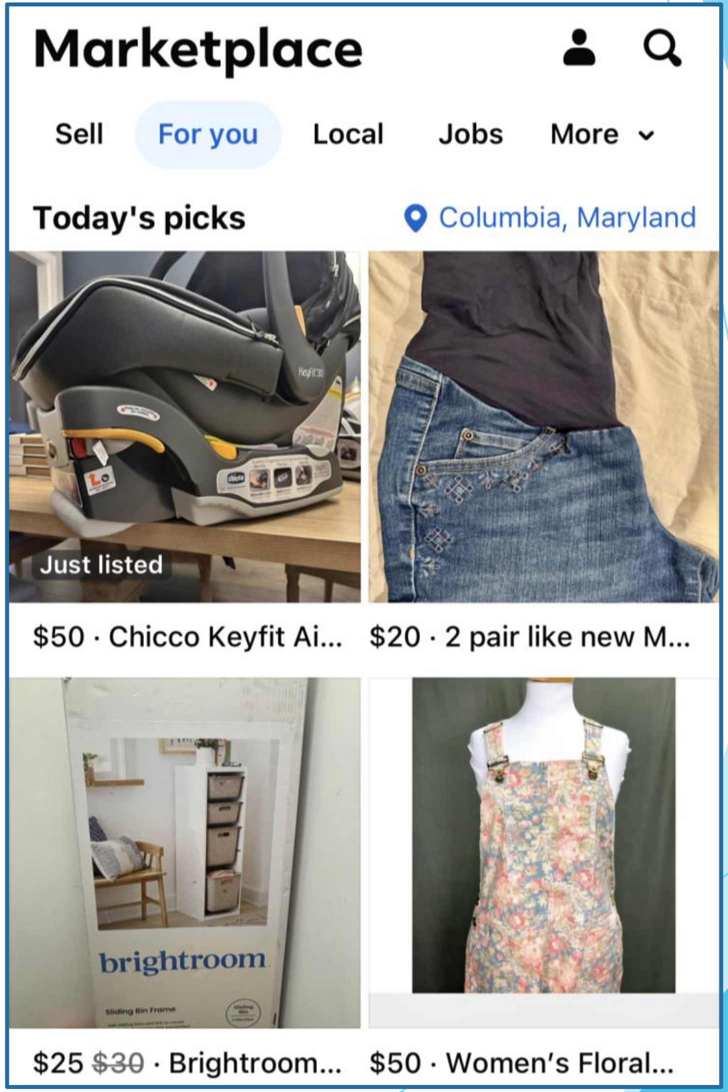
- Maintain short term memory and allow the user to make efficient references to that memory
- E.g., ChatGPT can modify previous queries





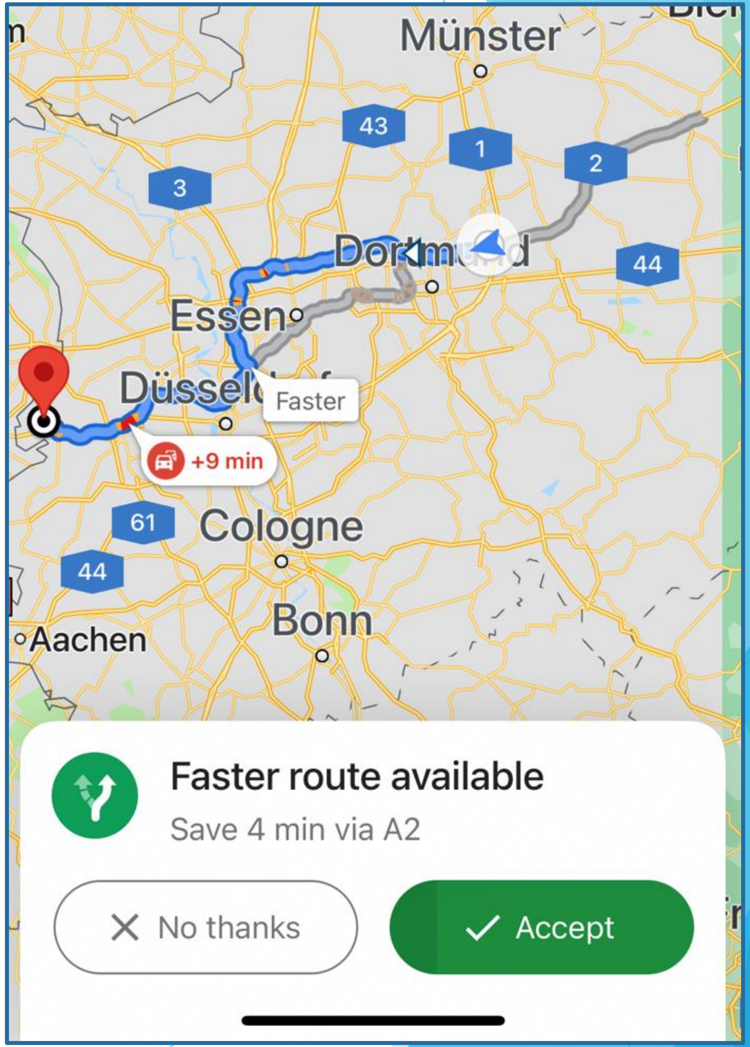
Learn from user behavior

- Personalize the user's experience by learning from their actions over time
- E.g., Facebook marketplace tailored to past searches



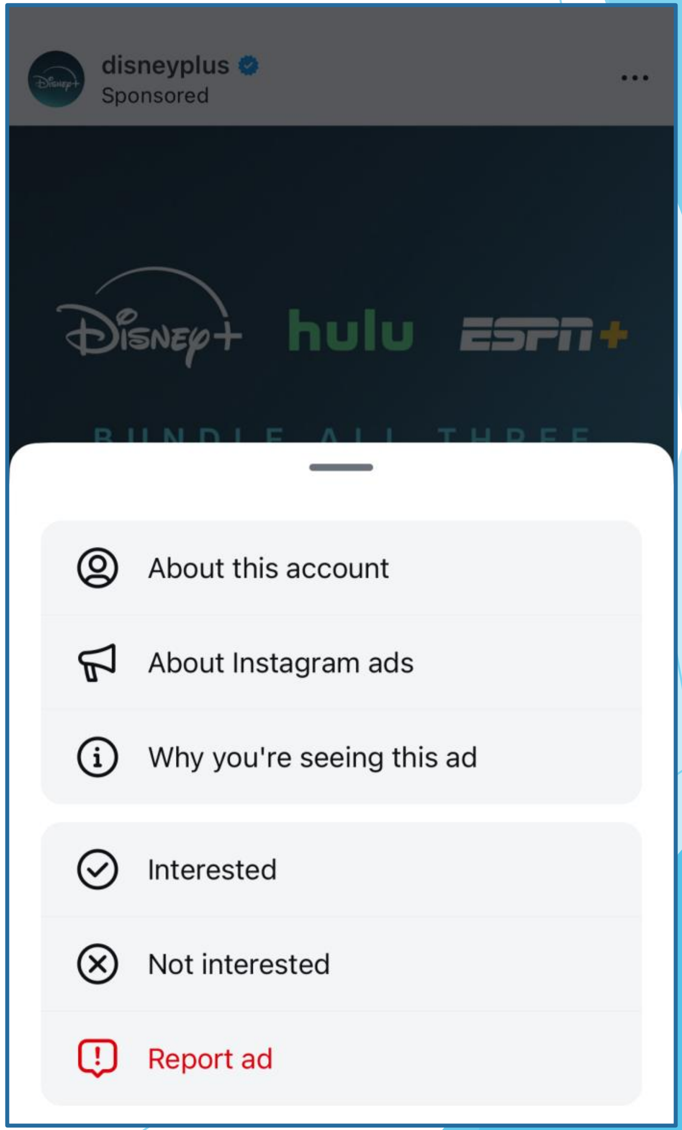
Update and adapt cautiously

- Limit disruptive changes when updating and adapting the AI system's behaviors
- E.g., Faster route notification



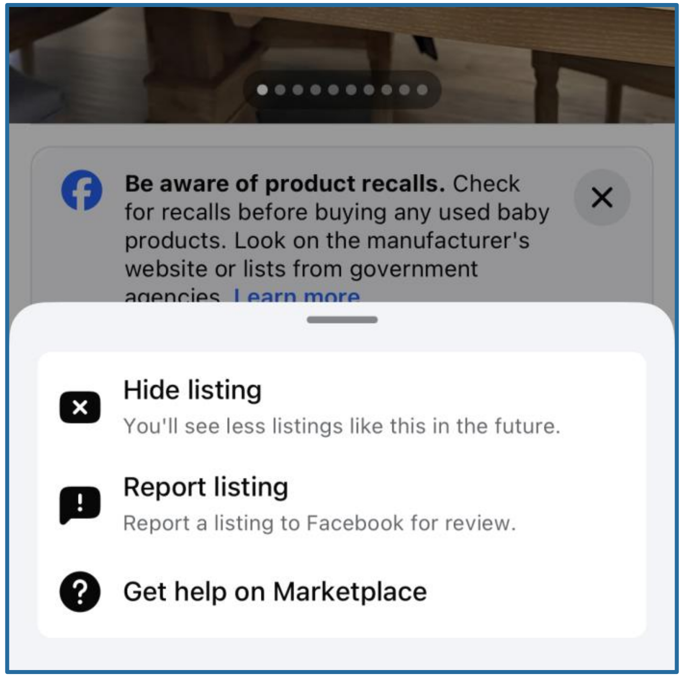
Encourage granular feedback

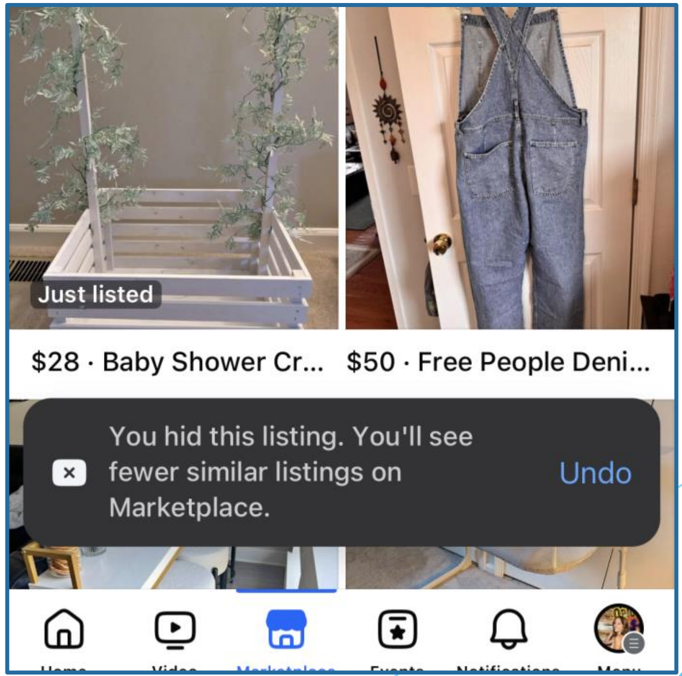
- Enable the user to provide feedback indicating their preferences during regular interaction with the AI system
- E.g., User can mark interested or not interested



Convey the consequences of user actions

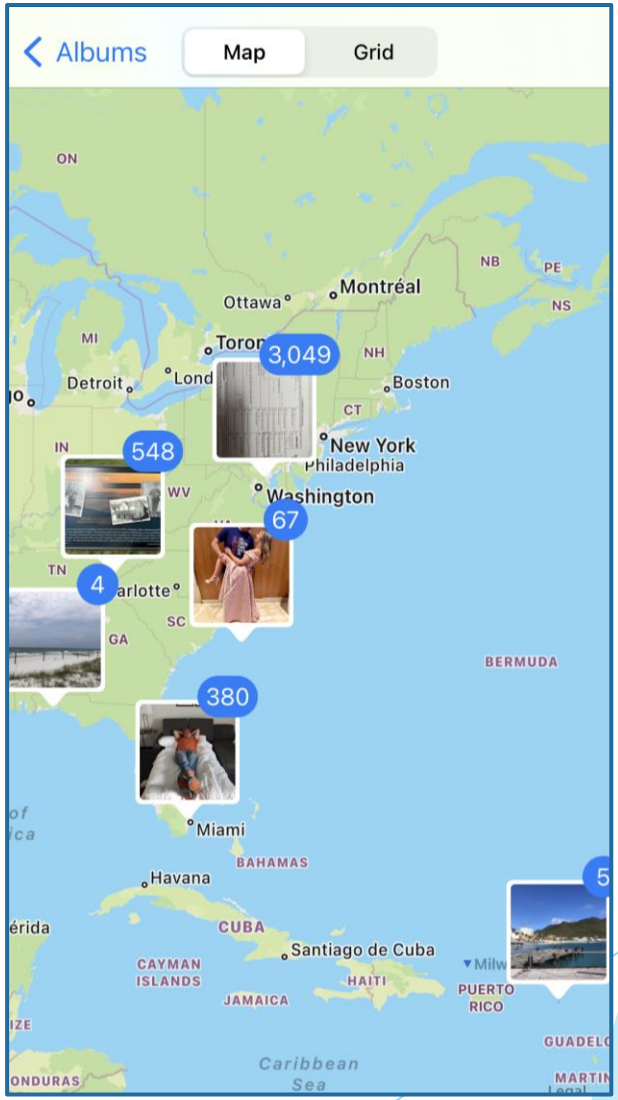
- Immediately update or convey how user actions will impact future behaviors of the AI system
- E.g., Facebook Marketplace message that will see fewer similar listings





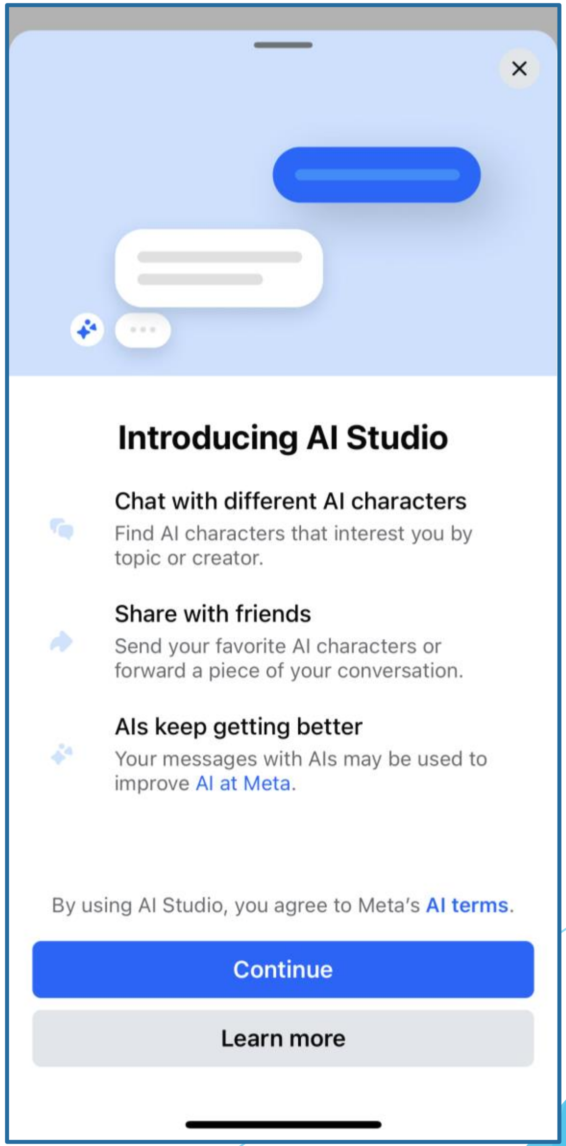
Provide global controls

- Allow the user to globally customize what the AI system monitors and how it behaves
- E.g., Enable location history to group photos



Notify users about changes

- Inform the user when the AI system adds or updates its capabilities
- E.g., Facebook AI Studio



Thank you!

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